

Galaxy Number Counts from a Deep Optical CCD Image

Kai J. Rogers

Abstract— This experiment measured the cumulative galaxy number–magnitude relation from a CCD FITS image and compared it with the Euclidean prediction. Sources were detected using Gaussian background fitting and sigma-threshold radial expansion, followed by adaptive aperture photometry with local background subtraction. The relation $\log_{10}N(<m)$ was linear in the bright regime, yielding a slope of 0.307 ± 0.004 per magnitude before flattening due to incompleteness. The measured slope differs significantly from the theoretical value of 0.6, which is likely caused by systematic effects.

I. INTRODUCTION

Counting galaxies as a function of apparent brightness provides information about the distribution of matter in the Universe [1].

The aim of this experiment is to determine the relationship between the cumulative number of galaxies brighter than magnitude m per unit sky area, and to compare this relation with the Euclidean prediction. This will be done by analysing a CCD FITS image. The success of the experiment is defined by obtaining a number-magnitude relation and identifying any differences from the theoretical prediction. 2 research questions which are answered through this experiment are:

1. What is the optimal sigma threshold above the background mean ($\mu + n\sigma$) for seed detection?
2. What is the measured gradient of the linear region of $\log_{10} N(<m)$ versus magnitude (where N is the number of sources with a magnitude less than m)?

II. THEORY

A central aim of a galaxy survey is to investigate how the galaxy population evolves over cosmic time. Such surveys involve deep observations of a “blank” region of sky. One way to study the galaxy population is through galaxy number counts, which measure how the number of detected galaxies varies with apparent brightness [2].

A Charge-Coupled Device (CCD) is an electronic detector that converts incoming photons into electrons within individual pixels, producing a digital image in which pixel counts are proportional to incident light. Each pixel has a finite charge capacity, so very bright sources can become non-linear or saturated. When pixels are overfilled, charge can spill into neighbouring pixels, producing vertical or horizontal streaks known as “blooming” [2]. The final image analysed in this experiment is created by combining multiple sub-exposures, which can result in increased noise near the edges where fewer exposures overlap, known as “edge-effects”. Light from astronomical point sources is blurred by atmospheric turbulence, known as “seeing”, which spreads the light over multiple CCD pixels rather than a single point. The Kitt Peak 4m telescope whose data is used in this experiment produces a point spread function of approximately 4 pixels [2].

Under the assumptions that galaxies are uniformly distributed and obey the inverse-square law for flux, the cumulative number of sources brighter than magnitude m is expected to follow the Euclidean relation [3]:

$$\log N(<m) = 0.6m + \text{constant}, \quad (1)$$

where N is the number of sources of a magnitude less than m .

III. EXPERIMENTAL METHOD

To exclude the blooming and edge effects of the CCD FITS image, the image was split into sections (tiles) in a way which avoided the edge, and bright stars that were blooming (Dimensions of tiles are specified in the appendix [B]). 2 stars were categorised to be blooming by visual inspection. The tile sections used are visualised in Fig.1.

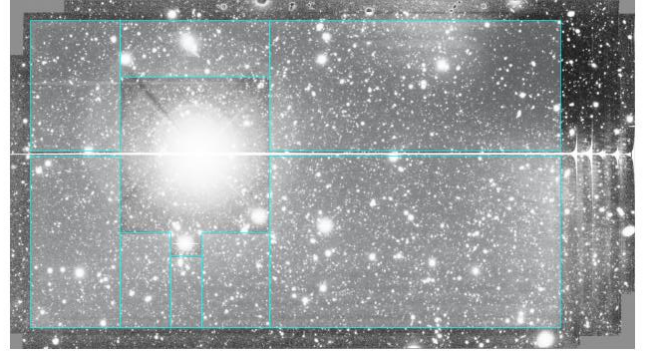


Figure 1. An Image Highlighting the tiles used in this experiment in blue. A bright star to the left can be seen blooming across the image.

The background mean was found by plotting a histogram of the values and fitting a Gaussian curve, an example is shown in Fig.2. The tail on the right-hand side of the gaussian curve is due to the sources.

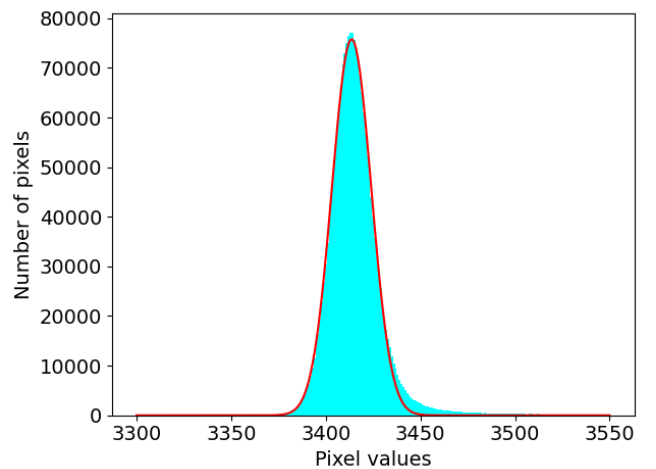


Figure 2. A histogram showing a gaussian fit to the pixel values of the top right tile in Fig.1. Notice the tail to the right of the curve.

A. Source Detection

For each tile, we detected sources by finding “seeds”, pixels brighter than $\mu + 3.0\sigma$ of their respective Gaussian curves. For each candidate source, radial expansion was performed by measuring the median pixel value in successive rings of 1 pixel thickness. We chose rings with 1 pixel thickness, as we wanted a thickness as small as possible, so that the median would update immediately every time the ring expanded, but we didn’t go lower than a 1 pixel width as we wanted the ring to be a full circle, as we didn’t want some directions to be bias over others. Expansion stopped when the ring median fell below $\mu + 1.25\sigma$, defining a source-specific radius. We excluded objects detected with a radius of less than 2 pixel because even point sources should have a greater radius due to seeing. We multiplied each found radius by a factor of $k=1.5$ to capture the wings of ellipse shaped galaxies.

The threshold for seed detection, was justified by testing different thresholds. After detection we removed the detected objects from the data, and repeated the gaussian fit for different seed thresholds (constant ring threshold). We found that $(3.0 \pm 0.05)\sigma$ resulted in the lowest reduced chi squared value. The number of standard deviations for the threshold of the median of the ring was chosen to balance the trade-offs. A large threshold results in the source being truncated; however a low threshold results in the risk of other nearby sources being included in the radius of the source. The value 1.25 was determined through visual inspection of what threshold values balanced these the best.

B. Photometry

Aperture photometry was performed using the measured radius for each source. Local background was estimated from an annulus spanning from the radius of the detected source to $radius + 15$ pixels. Net count from each source was calculated as the aperture sum minus the scaled background contribution. This value was then divided by 720 which is the time taken for the image in seconds, which gave us the flux. Instrumental magnitudes were converted to calibrated magnitudes using the image zero-point. At this point all sources from different tiles were combined into a single catalogue.

The annulus thickness of 15 pixels was chosen as a compromise between statistical precision and locality. Increasing the annulus width increases the number of pixels used in the background estimate, reducing its statistical uncertainty, but at the expense of sampling being less local. Varying the annulus thickness between 5 and 50 pixels produced negligible changes in the measured fluxes for sources, indicating that 15 pixels lies within a stable regime.

C. Number Counts and Fitting

Cumulative counts $\log N(< m)$ were calculated and normalised by survey area. Poisson uncertainties were propagated into log space using

$$\sigma_{\log 10} = \frac{1}{\sqrt{N} \ln 10} \quad [4](2)$$

A weighted linear regression was performed. We excluded magnitudes greater than the threshold at which they start to plateau as can be seen in Fig.3. As progressively fainter flux limits are reached, sources become increasingly difficult to detect, leading to incompleteness in the survey. This causes

the cumulative number counts to deviate from the expected linear relation and gradually flatten, as commonly observed in deep optical surveys [1].

IV. RESULTS

The optimal seed detection threshold was found to be $\mu + 3.0\sigma$. The cumulative number–count relation exhibits an approximately linear trend in the bright magnitude regime before flattening at faint magnitudes due to incompleteness. We obtained a final value of the lines gradient to be $s = (0.307 \pm 0.004)$ per mag (uncertainty only includes statistical uncertainty, not systematic).

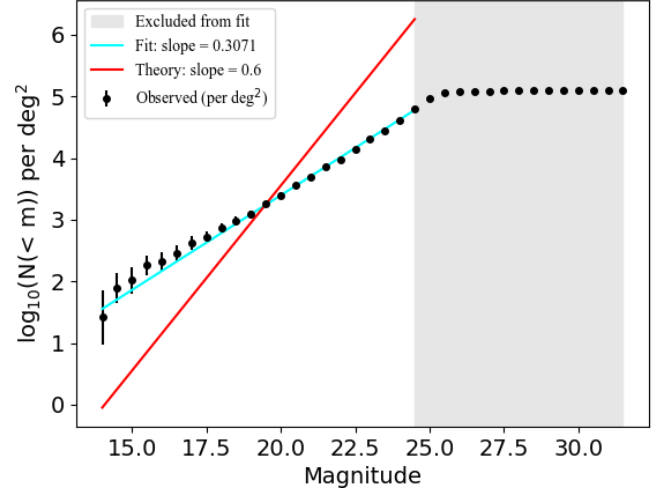


Fig.3. $\log_{10} N(< m)$ per degree² plotted against the magnitude. A line fit up to a magnitude of 24.5 is shown. The theoretical line is also shown. Notice how the data plateaus.

My results differ from the theoretical prediction by $\Delta s = (0.293 \pm 0.004)$ per mag. This corresponds to a $\sim 70\sigma$ deviation from the Euclidean prediction. Given the very small quoted statistical uncertainty, the discrepancy cannot be explained by counting statistics alone and therefore indicates the presence of significant systematic effects. The Euclidean prediction assumes a complete, flux-limited sample in a static, homogeneous, non-evolving universe [3]. While deviations from these cosmological assumptions can alter the expected slope, the magnitude of the observed difference suggests that methodological and survey-related effects are likely dominant. In particular, the use of fixed sigma thresholds for source detection introduces incompleteness at faint magnitudes, where low signal-to-noise sources may fail either the seed or radial expansion criteria. This leads to an underestimation of the cumulative counts at larger magnitudes and a departure from the theoretical relation.

V. CONCLUSIONS

Galaxy number counts were measured from a CCD FITS image using Gaussian background estimation, sigma-threshold source detection, adaptive aperture photometry, and weighted linear regression. The optimal seed detection threshold was found to be $\mu + 3.0\sigma$. The cumulative relation $\log_{10} N(< m)$ was approximately linear in the bright-magnitude regime, with a measured slope of $s = 0.307 \pm 0.004$ per magnitude before flattening due to incompleteness. The measured slope differs significantly from the Euclidean prediction of 0.6, indicating systematic effects in analysis methods.

REFERENCES

- [1] M. Yasuda et al., “Galaxy number counts from the Sloan Digital Sky Survey commissioning data,” *Astron. J.*, vol. 122, pp. 1104–1124, Sept. 2001
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- [4] J. R. Taylor, *An Introduction to Error Analysis: The Study of Uncertainties in Physical Measurements*, 2nd ed. Sausalito, CA, USA: University Science Books, 1997

APPENDIX

My previous report did not sufficiently justify the methodological choices. In this report, I quantitatively justified the seed detection threshold ($\mu + 3.0\sigma$) by varying the threshold value and evaluating its effect on the reduced χ^2 .

[A] Derivation of Equation (1):
Starting off with

$$N(< r) \propto r^3,$$

where N is the number of sources within radius r , and

$$f = \frac{L}{4\pi r^2},$$

where f is the flux of the source and L is the luminosity. These 2 equations can be combined or obtain a number–flux relation:

$$N(< f) \propto f^{-\frac{3}{2}}.$$

Magnitude definition:

$$\begin{aligned} m &= -2.5\log_{10}(f) + C \\ \log_{10}(f) &= -0.4m + C' \\ f &\propto 10^{-0.4m} \end{aligned}$$

Substitute into number–flux relation:

$$\begin{aligned} N(< m) &\propto (10^{-0.4m})^{-3/2} \\ N(< m) &\propto 10^{0.6m} \end{aligned}$$

which is equivalent to equation (1).

[B] Dimensions of tiles:

(y min, y max, x min, x max)
(3800, 4461, 150, 1420)
(3800, 4461, 1455, 2420)
(3425, 3800, 150, 850)
(3200, 3425, 150, 680)
(2700, 3200, 150, 850)
(2700, 3800, 2000, 2420)
(550, 2700, 150, 1420)
(550, 2700, 1455, 2420)

[C] Derivation of Equation (2) :

For Poisson counting statistics:

$$\sigma_N = \sqrt{N}$$

For the plotted quantity:

$$y = \log_{10}(N)$$

Using error propagation:

$$\sigma_y = \left| \frac{dy}{dN} \right| \sigma_N$$

Since

$$\begin{aligned} \log_{10}(N) &= \frac{\ln N}{\ln 10} \\ \frac{dy}{dN} &= \frac{1}{N \ln 10} \end{aligned}$$

Therefore,

$$\sigma_{\log_{10} N} = \frac{1}{N \ln 10} \sqrt{N} = \frac{1}{\sqrt{N} \ln 10}$$